

LIENZ, Austria

WMO: 112040

Lat: 46.826N

Lon: 12.806E

Elev: 668

StdP: 93.55

Time Zone: 1.00 (EUC)

Period: 94-19

WBAN: 99999

Annual Heating, Humidification, and Ventilation Design Conditions

Coldest Month	Heating DB		Humidification DP/MCDB and HR						Coldest Month WS/MCDB				MCWS/PCWD to 99.6% DB		WSF
			99.6%			99%			0.4%		1%				
	99.6%	99%	DP	HR	MCDB	DP	HR	MCDB	WS	MCDB	WS	MCDB	MCWS	PCWD	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
1	-13.8	-11.7	-15.8	1.0	-11.6	-14.2	1.2	-9.1	8.1	3.4	7.0	3.0	0.7	280	0.510

Annual Cooling, Dehumidification, and Enthalpy Design Conditions

Hottest Month	Hottest Month DB Range	Cooling DB/MCWB						Evaporation WB/MCDB						MCWS/PCWD to 0.4% DB	
		0.4%		1%		2%		0.4%		1%		2%			
		DB	MCWB	DB	MCWB	DB	MCWB	WB	MCDB	WB	MCDB	WB	MCDB	MCWS	PCWD
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)
7	11.9	30.0	20.0	28.2	19.2	26.5	18.3	20.8	28.5	19.8	26.8	18.9	25.2	4.1	120

Dehumidification DP/MCDB and HR									Enthalpy/MCDB						Extreme Max WB
0.4%			1%			2%			0.4%		1%		2%		
DP	HR	MCDB	DP	HR	MCDB	DP	HR	MCDB	Enth	MCDB	Enth	MCDB	Enth	MCDB	
(a)	(b)	(c)	(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	
18.0	14.0	24.8	17.1	13.3	23.2	16.4	12.7	21.9	63.1	28.7	59.4	26.8	56.3	25.3	23.8

Extreme Annual Design Conditions

Extreme Annual WS			Extreme Annual Temperature				n-Year Return Period Values of Extreme Temperature									
			Mean		Standard Deviation		n=5 years		n=10 years		n=20 years		n=50 years			
1%	2.5%	5%	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	Min	Max	DB	WB
(a)	(b)	(c)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(4) 7.5	6.3	5.3	-16.2	32.7	2.6	2.0	-18.1	34.1	-19.6	35.3	-21.0	36.4	-22.9	37.8		
(5)			-16.4	22.2	2.5	1.3	-18.3	23.1	-19.7	23.9	-21.2	24.6	-23.0	25.5	(4)	(5)

Monthly Climatic Design Conditions

		Annual	Jan	Feb	Mar	Apr	May	Jun	Jul	Aug	Sep	Oct	Nov	Dec		
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)	(n)	(o)	(p)		
(6) Temperatures, Degree-Days and Degree-Hours	DBAvg	8.9	-3.0	0.1	5.1	9.5	14.0	17.7	19.3	18.8	14.2	9.3	3.1	-2.1		
	DBStd	8.53	4.10	3.75	3.68	3.41	3.09	3.46	2.91	2.92	3.03	3.44	3.77	3.71		
	HDD10.0	1532	404	277	156	49	4	1	0	0	3	54	208	376		
	HDD18.3	3592	662	511	411	266	139	53	24	30	127	279	457	634		
	CDD10.0	1123	0	0	3	34	128	231	289	273	129	34	1	0		
	CDD18.3	140	0	0	0	0	4	33	54	45	3	0	0	0		
	CDH23.3	1317	0	0	0	4	69	349	482	377	35	0	0	0		
	CDH26.7	341	0	0	0	0	10	97	131	100	2	0	0	0		
(14) Wind	WSAvg	1.7	1.3	1.5	1.9	2.2	2.2	2.2	2.1	1.8	1.5	1.3	1.2	1.2		
(15) Precipitation	PrecAvg	1250	64	62	73	87	109	134	140	138	108	126	130	79		
	PrecMax	1603	219	204	158	153	218	211	264	213	191	339	389	189		
	PrecMin	1009	14	20	14	25	40	72	42	69	43	8	7	12		
	PrecStd	163	41	38	36	31	40	37	48	36	38	77	94	37		
(19) Monthly Design Dry Bulb and Mean Coincident Wet Bulb Temperatures	0.4%	DB	10.2	13.5	19.8	23.3	28.2	31.9	32.2	32.2	26.7	21.3	13.6	8.3		
		MCWB	5.6	7.4	10.5	13.8	17.8	20.0	21.0	20.6	18.8	14.9	9.2	4.2		
	2%	DB	6.2	10.1	17.0	21.0	25.3	29.4	29.8	29.2	24.1	19.0	11.5	5.6		
		MCWB	2.4	5.0	9.4	12.8	16.2	19.3	20.3	20.0	17.1	13.4	8.7	3.1		
	5%	DB	3.8	7.7	14.6	19.0	23.1	27.4	28.0	27.2	22.1	17.2	10.1	3.9		
		MCWB	1.5	3.5	8.2	11.6	15.0	18.2	19.2	19.2	15.9	12.4	8.2	2.2		
	10%	DB	2.2	5.7	12.2	16.8	21.1	25.2	26.1	25.2	20.3	15.3	8.7	2.5		
		MCWB	0.5	2.4	6.8	10.3	13.9	17.1	18.3	18.3	15.1	11.6	7.1	1.2		
(27) Monthly Design Wet Bulb and Mean Coincident Dry Bulb Temperatures	0.4%	WB	6.1	7.6	11.4	14.6	18.5	21.5	22.2	22.0	19.1	15.5	10.6	5.5		
		MCDB	9.9	12.9	18.5	22.6	26.6	29.6	30.7	30.4	25.8	20.1	12.0	6.9		
	2%	WB	3.1	5.4	9.8	13.1	16.7	20.0	20.9	20.6	17.6	14.1	9.5	3.6		
		MCDB	5.0	9.4	16.2	20.0	24.0	27.6	28.6	28.1	23.0	17.6	10.8	4.9		
	5%	WB	1.9	3.9	8.5	11.9	15.6	18.9	19.8	19.5	16.5	13.1	8.5	2.4		
		MCDB	3.3	7.0	14.0	17.8	22.0	26.0	26.6	26.2	21.0	16.1	9.8	3.6		
	10%	WB	0.8	2.7	7.1	10.7	14.4	17.8	18.8	18.6	15.6	12.0	7.1	1.4		
		MCDB	2.0	5.2	11.5	15.9	19.8	23.9	24.9	24.3	19.4	14.9	8.4	2.3		
(35) Mean Daily Temperature Range	5% DB	MDBR	8.0	9.6	11.6	11.3	11.5	11.7	11.9	11.3	10.9	9.9	7.4	6.8		
		MCDBR	9.3	12.9	16.2	15.6	15.6	15.7	14.9	14.8	13.8	12.6	9.3	7.9		
	5% WB	MCWBR	6.5	8.1	9.1	8.1	7.4	6.8	6.5	6.5	7.3	7.4	6.2	5.8		
		MCWBR	8.2	12.0	15.4	14.4	14.4	14.3	14.0	13.6	12.2	11.1	7.5	6.9		
(40) Clear-Sky Solar Irradiance	taub		0.292	0.328	0.389	0.450	0.467	0.478	0.469	0.461	0.416	0.383	0.336	0.295		
		taud	2.297	2.183	2.081	2.017	2.043	2.072	2.112	2.148	2.239	2.282	2.348	2.367		
	Ebn at Noon		790	824	818	802	804	795	797	785	787	754	728	745		
		Edh at Noon	76	106	137	161	163	159	151	140	117	96	74	65		
(44) All-Sky Solar Radiation	RadAvg		1.43	2.30	3.38	4.20	4.88	5.46	5.57	4.89	3.64	2.45	1.42	1.20		
	RadStd		0.16	0.30	0.41	0.55	0.60	0.47	0.45	0.51	0.44	0.28	0.26	0.17		

Historical Trends

		DBAvg	Heating		Cooling			Degree-Days			
			99% DB	99% DP	1% DB	1% WB	1% DP	HDD10.0	HDD18.3	CDD10.0	CDD18.3
		(d)	(e)	(f)	(g)	(h)	(i)	(j)	(k)	(l)	(m)
(46) Station Only		+0.84	+1.58	N/A	+1.33	N/A	N/A	-218	-269	+147	+79
(47) Regional (81 neighbors)		+0.43	N/A	N/A	+0.63	+0.40	+0.35	N/A	N/A	+74	+38

Nomenclature: See separate page